

Jorge Leal

Fullstack Developer

Mexico

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Professional Summary

Fullstack Developer focused on web platforms, high-performance backend systems, and data-driven dashboards. Experience building multi-tenant SaaS products, hybrid (SSG/SSR) architectures, headless CMS solutions, and performance-optimized services leveraging advanced caching strategies. Hands-on experience designing LLM-powered tools, multi-agent orchestration systems, and clean architectures for development workflows and automation.

Frontend:

Next.js 14, React 18, SvelteKit, Tailwind CSS, NiceGUI, Astro, Three.js

Backend:

Rust (Axum), FastAPI, Go (Fiber), KeystoneJS, Node.js, Python

Databases:

PostgreSQL (Neon), SQLite, Redis

AI / LLM Systems:

OpenAI, Groq, Gemini, MusicGen, VS Code Language Model API, MCPPayments:

DevOps / Deploy:

Docker, Fly.io, Cloudflare, GitHub Codespaces

Architecture & Tooling:

Clean Architecture, Ports & Adapters, Monorepos (pnpm workspaces), Event-Driven Systems AI / Data / Visualization

Media & Performance:

Cloudinary, proprietary image CDN (PixMinder)

Others:

Stripe, GraphQL (URQL), Axios, Plotly

Projects

Agent System | TypeScript (ES2022), VS Code API, SQLite, LLM Orchestration

VS Code extension implementing a project-aware multi-agent coding assistant with persistent memory, optional RAG, and secure tool execution integrated into the Sidebar and Copilot Chat (@agent).

Built with Clean Architecture in a pnpm monorepo: decoupled core (EventBus, orchestration, interfaces), infra adapters, and app activation layer.

Specialized agents (Coder, Reviewer, Tester, Coordinator) coordinated via hybrid routing (keywords + embeddings + LLM), TTL caching, collaborative orchestration, and Auto/Agent/Plan modes with configurable reasoning levels.

Advanced memory & RAG: persistent SQLite memory (session/project/global) with TTL, optional OpenAI embeddings, hash-gated indexes, and static workspace analysis (AI Project Awareness / Project Context).

Secure tool management & LLM fallback: read/edit/create/delete files, run_tests, and search_web with risk-based approval; chained LLMs (Copilot → VS Code LM → MCP fallback); configurable web search provider.

PixMinder | Rust (Axum), PostgreSQL, Redis

High-performance image optimization service designed as a CDN alternative to replace Cloudinary native transformations.

Backend implemented in Rust using Axum, following Clean / Hexagonal Architecture with strict layer separation.

Context-aware image optimization with automatic format selection (AVIF, WebP, JPEG, PNG), adaptive quality, DPR-aware resizing, and SVG pass-through.

Multi-layer caching architecture (in-memory, PostgreSQL, Redis) achieving sub-10ms cache latency and 15–30% better compression than Cloudinary native.

Production usage serving Qubly CRM and LaZalza, delivering +7–13 Lighthouse Performance points and 300–600ms LCP reduction.

Databoard | FastAPI, NiceGUI, Plotly, PostgreSQL (Neon)

Multi-tenant analytics dashboard for real-time monitoring of web metrics across multiple sites and projects.

Async architecture with tenant-level data isolation, optimized caching, and resilient error handling. Advanced Plotly visualizations including time-series charts, hourly heatmaps, geolocation maps, and Spotify-style KPIs.

Key features include multi-site selection, audience breakdown, time zone adjustment, CSV export, and statistical forecasting (EWMA, trend analysis).

Qubly CRM | SvelteKit, Go (Fiber), PostgreSQL (Neon)

Multi-tenant SaaS platform for small businesses in Mexico to launch an online store in minutes with direct WhatsApp ordering and zero sales commissions.

Hybrid architecture with SvelteKit (SSG, SSR, PWA) frontend and Go backend following Clean Architecture principles.

Full e-commerce system with recurring subscriptions, Stripe Connect, webhooks, and a built-in referral system.

AI-powered chatbot using Groq (Llama) optimized for low cost via intent detection, efficient context construction, and response caching.

Kidyboard (Private) | SvelteKit, FastAPI, PostgreSQL (Neon)

Multi-role management system for children's entertainment centers with administrative dashboard, sales, inventory, services, and real-time reporting.

Hybrid SvelteKit frontend with monorepo architecture and FastAPI backend following Clean Architecture.

Operational dashboard with real-time metrics (sales, ATV, inventory, active timers), comparative reports, and statistical forecasting for sales and capacity planning.

Advanced features include real-time timers, Excel/PDF report exports, digital menu generation (PDF and video), and granular role-based access control.

Beat Catalogue | SvelteKit, FastAPI, PostgreSQL

Music catalog platform with hybrid SSR and Spotify-style modal overlay experience. FastAPI backend with PostgreSQL and custom admin panel.

Audio player with shuffle mode, play counter, and persistent theme switcher.

LaZalza | Next.js, KeystoneJS, PostgreSQL

News and editorial website focused on music, cinema, and popular culture.

Frontend built with Next.js 14 App Router and ISR (5-minute revalidation).

Headless CMS using KeystoneJS with Cloudinary integration and interactive 3D logo.

Raccoonin Records | Next.js, KeystoneJS, PostgreSQL

Platform for an independent music label with hybrid architecture.

Migration from CSR to ISR with infinite post pagination.

Multimedia embeds, streaming platform links, and dynamic sitemap generation.

BSTCGenerator | FastAPI, MusicGen, Gemini

Audio generation platform for music producers with cost-efficient architecture.

Decoupled FastAPI backend with AI worker running on Google Colab (GPU) using MusicGen.

Embedding and recommendation system based on user feedback and pattern analysis.

Education

Universidad Abierta y a Distancia de México

Bachelor in Software Engineering January 2025 – Present

Universidad Autónoma Metropolitana, Unidad Xochimilco

Bachelor in Social Communication Graduated: October 2015

Languages

Spanish: Native English: Intermediate